

POINT SET

Dr. Joshua Kiddy Kwasi Asamoah

Department of Mathematics
Kwame Nkrumah University of Science and Technology



Lecture Outline

- 1 Intervals
 - Open and Closed Intervals
 - Bounded and Unbounded Intervals
- 2 Intervals and Absolute Value
- 3 Neighbourhoods
 - δ -Neighbourhood
 - Deleted Neighbourhood
- 4 Isolated Point, and Limit Point
- 5 Bound Sets



Outline of Presentation

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Open intervals

Let a and b be two real numbers such that $a \leq b$.

Definition

The set $I = \{ x \in \mathbb{R} \mid a < x < b \}$ is called an **open interval** and is denoted by $I = (a, b)$



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Example

$(-1, 2)$ is an open interval and does not include its endpoints -1 and 2 .



Closed Interval

Let a and b be two real numbers such that $a \leq b$.

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The set $I = \{ x \in \mathbb{R} \mid a \leq x \leq b \}$ is called a **closed interval** and is denoted by $I = [a, b]$



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Example

$[\frac{2}{3}, 2\sqrt{3}]$ is a closed interval and **does include its endpoints** $\frac{2}{3}$ and $2\sqrt{3}$.



Bounded intervals: Half open or half closed intervals

Definition (Left-closed and right-open)

The set $[a, b) = \{ x \in \mathbb{R} \mid a \leq x < b \}$ is a **left-closed and right-open interval**.



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Definition (Bounded intervals)

The intervals (a, b) , $(a, b]$, $[a, b)$ and $[a, b]$ are called **bounded intervals**.
 a is the **greatest lower bound** and b the **least upper bound**.



Unbounded intervals: Left open or closed intervals

Definition (Left-open)

The set $(a, +\infty) = \{ x \in \mathbb{R} \mid a < x \}$ is a **left-open interval**.



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Definition (Unbounded Interval)

The set $(-\infty, +\infty)$ is the set of real numbers, $\mathbb{R}$. It's **unbounded below** and **above**.



Outline of Presentation

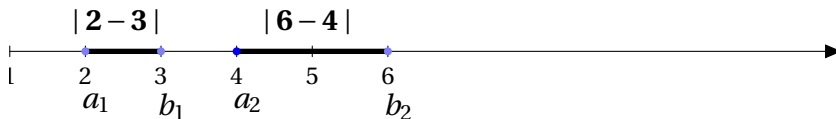
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Intervals and absolute value

Geometric interpretation

The absolute value between two real numbers a and b is the distance between them.



Intervals and absolute value

Geometric interpretation

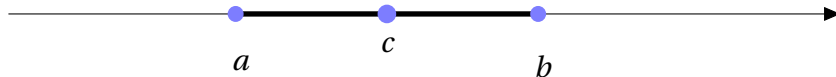
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Intervals and absolute value

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It admits a midpoint and a radius.

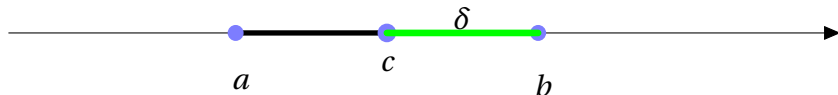
Definition (Midpoint and Radius)

The **midpoint** of I is the real number $c = \frac{1}{2}(b + a)$.

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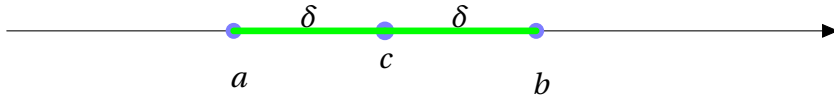
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The **midpoint** of I is the real number $c = \frac{1}{2}(b + a)$.

The **radius** of I is the real number $\delta = \frac{1}{2}(b - a)$.

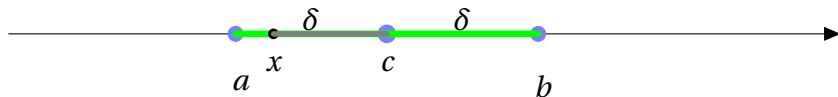
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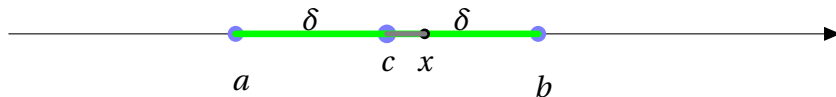


The *distance* from any point x in the interval to the *midpoint* is **less than or equal to** the *radius*.



Intervals and absolute value

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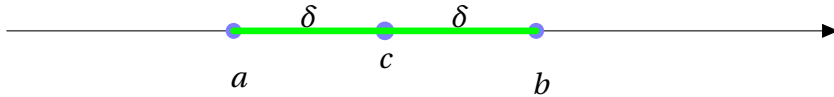


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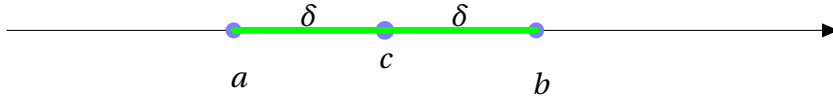
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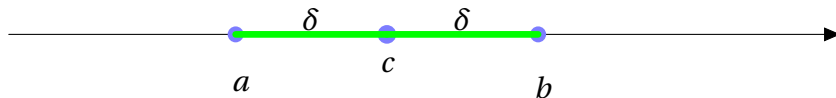
Property

$$\textcircled{1} \quad (a, b) = \{ x \in \mathbb{R} \mid a < x < b \} = \{ x \in \mathbb{R} \mid |x - c| < \delta \}.$$



Intervals and absolute value

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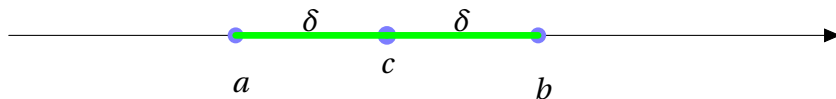
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Intervals and absolute value

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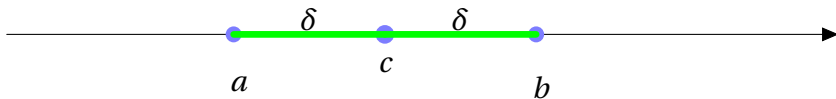
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- 3 $(-\infty, a) \cup (b, \infty) = \{ x \in \mathbb{R} \mid x < a \text{ or } b < x \} = \{ x \in \mathbb{R} \mid |x - c| > \delta \}.$



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Remark

$a = c - \delta$ and $b = c + \delta$.

Example

Solve $|x - 8| \leq 2$.



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① $|3 - 2x| > 7$ implies $|-2(x - 3/2)| > 7$ which implies $|x - 3/2| > 7/2$.



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- ② We have $c = 3/2$, $\delta = 7/2$, $a = c - \delta = -2$ and $b = c + \delta = 5$.
- ③ Since the inequality is $>$ the solution set is the interval $I = (-\infty, -2) \cup (5, +\infty)$.



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- 3 Because $|2(x + 2)| = 2|x + 2|$, we have $|x + 2| < 1$.



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- 5 Since the inequality is $<$ the solution set is the interval $I = (-3, -1)$.



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δ -Neighbourhood

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Let a be a point and $\delta > 0$. The open interval $(a-\delta, a+\delta)$ is called the **neighbourhood** (sometimes the **δ -neighbourhood**) of a that is, the set of all points x satisfying $a-\delta < x < a+\delta$.



Figure 1: δ -neighbourhood of a .



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Let $\mathbf{a}$ be a point and $\delta > 0$. The open interval $(a-\delta, a+\delta)$ is called the **neighbourhood** (sometimes the **δ -neighbourhood**) of $\mathbf{a}$ that is, the set of all points x satisfying $a - \delta < x < a + \delta$.

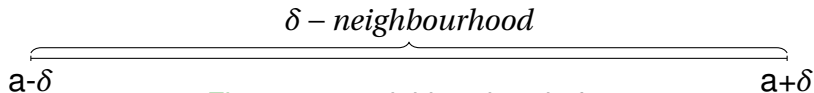


Figure 1: δ -neighbourhood of a .

Thus δ -neighbourhood of $\mathbf{a}$ is the open interval centered at $\mathbf{a}$ and has length 2δ .

Open and Closed δ -neighbourhoods

Note

$$|x - x_0| \leq \delta \quad (1)$$

$$-\delta < x - x_0 < \delta \quad (2)$$

$$x_0 - \delta < x < x_0 + \delta \quad (3)$$

Given $x_0 \in \mathbb{R}$ and $\delta > 0$,

Definition (Open δ -neighbourhood)

The open interval defined by $|x - x_0| < \delta$ is called the **open δ -neighbourhood** of x_0 .



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Definition (Closed δ -neighbourhood)

The closed interval defined by $|x - x_0| \leq \delta$ is called the **closed δ -neighbourhood** of x_0 .

Deleted neighbourhood

Definition (Deleted neighbourhood)

The set of all points x such that

$$0 < |x - x_0| < \delta \quad (4)$$

is called a deleted δ -neighbourhood of x_0 .

The deleted δ -neighbourhood of x_0 is an interval with centre x_0 and length 2δ , with the point x_0 excluded from the interval.



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- 1 The 1-neighbourhood of 0 is the open interval $(-1, 1)$. It has length 2 and centre 0.

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- 2 The deleted 2-neighbourhood of 0 is the interval $\{x : -2 < x < 2; x \neq 0\}$.

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Isolated points

Let S be a subset of $\mathbb{R}$ and x_0 an arbitrary real number.

Definition

x_0 is called an **isolated point** of S if for $x_0 \in S$, there exists a δ -neighbourhood of x_0 (I_δ) which **does not contain any other points** of S ; that is

$$I_\delta \cap S = \{x_0\} \quad (5)$$



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Example

- ① 0 is an isolated point of $S = \{0\} \cup [1, 2)$.
- ② Except 0, all the points of $S = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$ are isolated points.

Limit point: Let S be a subset of $\mathbb{R}$ and x_0 an arbitrary $\mathbb{R}$.

Definition (Limit Points)

x_0 is a **limit point** (or **cluster point** or **accumulation point**) of S if:

- 1 every δ -neighbourhood of x_0 contains **at least one point** of S **different** from x_0 itself.



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x_0 is a **limit point** (or **cluster point** or **accumulation point**) of S if:

- 1 every δ -neighbourhood of x_0 contains **at least one point** of S **different** from x_0 itself.
- 2 x_0 **does not itself have to be an element of S .**



Note

- 1 No finite set has a limit point

Example the set $\{1,2,3,4,5\}$ has no limit points since for any suggested number, a deleted δ -neighbourhood can be constructed with no elements of the set present.



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Example the set $\{1,2,3,4,5\}$ has no limit points since for any suggested number, a deleted δ -neighbourhood can be constructed with no elements of the set present.

- 2 An infinite set may or may not have a limit point. In particular, the natural numbers $N = \{1,2,3,\dots\}$ has no cluster points, but the set of rational numbers has infinitely many limit points.



Example

- 1 0 is a limit point of the interval $(0, 1)$ since every deleted δ -neighbourhood of 0 will contain some point in $(0, 1)$.

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- ② The set $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ has no limit point. In general no finite set of numbers has a limit point.
- ③ Every point in $[0, 1]$ is a limit point.
- ④ The set $\{2, 4, 6, 8, 10, \dots\}$ has no limit points since we can define some deleted δ -neighbourhood of the elements of the set which will contain no element of the set. In general an infinite set may or may not have limit point.
- ⑤ Elements of the interval $[1, 2]$ are limit points of $S = \{0\} \cup [1, 2)$.
- ⑥ Only 0 is a limit point of $S = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$.

Closed Set

Definition (Closed Set)

A set is said to be **closed** if it contains all its limit points.



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A set is said to be **closed** if it contains all its limit points.

Example

The set $[0, 1]$ is a closed set, while the set $(0, 1)$ is not a closed set since 0 and 1 even though are limit points do not belong to the set.



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Upper Bound

Definition (Upper Bound)

Let A be a set of numbers, if there is some $M \in \mathbb{R}$ such that $x \leq M$, $\forall x \in A$, then A is said to be **bounded above**.

M is called an **upper bound** of the set A .



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Definition (Least Upper Bound)

Let $\underline{M}$ be an upper bound of A . Then if for any $\epsilon > 0$ (however small), there exist at least one element $x \in A$ such that

$$x > \underline{M} - \epsilon \tag{6}$$

then the number $\underline{M}$ is called the **least upper bound (l.u.b.)** of A .

Lower Bound

Definition (Lower Bound)

Let A be a set of numbers, if there exist some $m \in \mathbb{R}$ such that $x \geq m$, $\forall x \in A$, then A is said to be **bounded below**.

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Lower Bound

Definition (Lower Bound)

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Definition (Greatest Lower Bound)

Let $\overline{m}$ be a lower bound of A , then if for any $\epsilon > 0$ (however small), there exist at least one element $x \in A$ such that

$$x < \overline{m} + \epsilon \quad (7)$$

the number $\overline{m}$ is called the **greatest lower bound (g.l.b.)**.

Bounded Set

Definition (Bounded Set)

Let A be a set of numbers, if $\forall x \in A$; there some $m \in \mathbb{R}$ and $M \in \mathbb{R}$ such that $m \leq x \leq M$, then the set A is said to be **bounded**.

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Theorem (Bolzano-Weierstrass theorem)

Every **bounded infinite** subset of $\mathbb{R}$ has at least one limit point.



Example

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- ② The least upper bound is 1
- ③ The greatest lower bound is 0.
- ④ Since A is infinite, it has at least one limit point. The limit point is 0.

Exercise

- 1 Let $I_1 = \{x \in \mathbb{R} \mid -4 < x \leq 4\}$ and $I_2 = [-2, 8)$.
 - 1 Find I , the intersection of I_1 and I_2 .
 - 2 Find the midpoint c and the radius δ of I .
 - 3 Use the absolute value inequality to define I in terms of δ and c .
- 2 Solve: a) $|4x - 3| < 4$ b) $|-2x - 10| > 6$. c) $|3x + 1| < 5$
 d) $|3 - 2x| > 1$ e) $|x^2 - 2x + 3| < -3$.
- 3 Show that $\{1, 1/2, \dots, 1/n, \dots\}$ is bounded and determine its greatest lower bound and least upper bound.
 Again, identify some limit points and how does it illustrate the Bolzano-Weierstrass theorem.
- 4 Consider the set $\{1, 1.1, 0.9, 1.01, .99, 1.001, .999, \dots\}$.
 - 1 Is the set bounded?
 - 2 Does it have a greatest lower bound and least upper bound?
 - 3 Determine its limit points and determine if it is closed.



END OF LECTURE
THANK YOU

